



The device was made of aluminum and plastic.

Artificial heart

- **What?** Jarvik 7
- **Who?** Robert Jarvik
- **Where and when?** US, 1982

In 1982, an American dentist suffering from heart failure became the first person to be fitted with an artificial heart—allowing him to live for another 112 days. Named after its inventor, the Jarvik 7 heart was the first of many to be created over the following decades. In some cases, an artificial heart may buy time for a patient until a real heart becomes available for transplant.

Prosthetic legs

- **What?** Flex-Foot
- **Who?** Van Philipps
- **Where and when?** US, 1996

Lightweight and flexible, modern artificial or prosthetic limbs enhance the mobility of the user. The American bioengineer Van Philipps, who had lost a leg below the knee at the age of 21, invented a range of prosthesis called Flex-Foot. Among these is the Flex-Foot Cheetah, which is made of a carbon-fiber “blade,” and is aimed at athletes. It flexes on impact with the ground, helping an athlete move quickly.

► BLADE RUNNERS

A long jumper competes in the 2011 World Para Athletics Championship. Athletes' prosthetic legs act like springs, allowing them to run and jump at high speeds.

Spikes on the bottom of the blade help to grip the track.



Bionic hand

- **What?** I-Limb
- **Who?** David Gow
- **Where and when?** UK, 2007

Invented at a Scottish hospital, the I-Limb was the world's first fully functioning bionic (electronically controlled) hand. Its fingers can move independently according to muscle signals sent from the patient's arm. Patrick Kane (right) was the first person to be fitted with the device, aged 13.



FAST FACTS

- Around the 10th century, the Chinese were using magnifying lenses to read.
- Hearing aids evolved out of telephone technology—transmitters that amplified the sound in telephones were adapted for use in early hearing aids.
- The British surgeon Sir John Charnley performed the first successful hip replacement operation in the early 1960s.